Asset Registry Management System – Project Documentation

Platform: Microsoft Power Platform (Power Apps, Dataverse, Power Automate) **Author:** Shravan
Submission deadline: 22 October 2026

Project Status

Status: Completed. All MVP deliverables were built and tested before the 22 October 2026 deadline.

Component	Delivered
Dataverse tables (Asset, Checkout) with one-to-many relationship	Completed
Canvas app: Dashboard, Check Out, Check In, Overdue screens	Completed
Flow_CheckoutAsset (server-side double check-out prevention)	Completed
24-hour reminder flow with Teams @mention	Completed
Power analytics (utilization by type, overdue trend)	Completed
Acceptance testing	Completed, all tests passed

1. Introduction

1.1 Business problem

Organizations lend out laptops and peripherals (mouse, keyboard, monitor, headset) to staff for short periods. Without a proper register it is hard to know which items are available, who holds what, and which items are overdue. The Asset Checkout System keeps a simple asset register and tracks every temporary check-out from issue to return.

1.2 Objectives

- Maintain a central register of assets with type, tag, condition and status.
- Record each check-out and check-in against a user with a due date.
- Prevent an asset from being checked out twice at the same time.
- Automatically remind the assignee 24 hours before the due date through Microsoft Teams.
- Provide analytics on asset utilization and overdue trends.

1.3 Core learning goals

- One-to-many relationships in Dataverse (one asset, many checkouts).
- Using `Patch()` for check-out and check-in.
- Scheduled reminders using Power Automate.

2. Scope

2.1 In scope (MVP)

- Two Dataverse tables: **Asset** and **Checkout**.
- A canvas app with a dashboard, Check Out, Check In and Overdue screens.
- A reminder flow (24 hours before due date, Teams mention to the assignee).
- Analytics: utilization by asset type and overdue trend.

2.2 Stretch goal

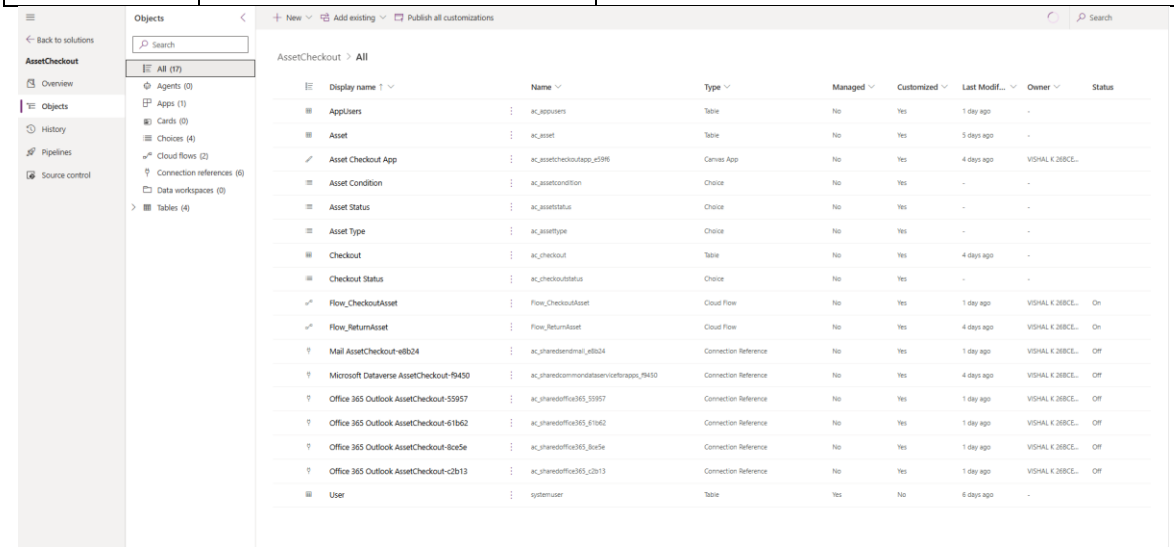
- AI Builder Object Detection to identify the asset type from a photo (learning-oriented, not required for acceptance).

2.3 Acceptance criteria

1. No asset can be checked out twice at the same time.
2. Overdue reminders fire correctly before the due date.

3. Solution Architecture

Layer	Component	Purpose
Data	Dataverse (Asset, Checkout)	Stores the register and the checkout history
App	Power Apps canvas app	User interface for staff/IT admin
Automation	Power Automate	Check-out flow and 24-hour reminder flow
Notification	Microsoft outlook	Reminder message with @mention
Analytics	Graph	Utilization and overdue-trend reports
AI Builder	object orientation	scanner integration doesn't exist, spec requirements needed.



4. Data Model

4.1 Asset table

Column	Type	Description
Name	Text (primary)	Friendly name, e.g. "Dell Latitude 5440"
Tag	Text	Unique asset tag / barcode value
Type	Choice	Laptop, Mouse, Keyboard, Monitor, Headset, Other
Condition	Choice	New, Good, Fair, Damaged
Status	Choice	Available, Checked Out, Under Repair, Retired

4.2 Checkout table

Column	Type	Description
Checkout ID	Auto number / primary	Unique reference
Asset	Lookup → Asset	The asset being borrowed
User	Lookup → User (or email)	The person who has the asset
From	Date and time	When the asset was issued
Due	Date and time	When it must be returned
Returned On	Date and time	Empty until the asset is returned

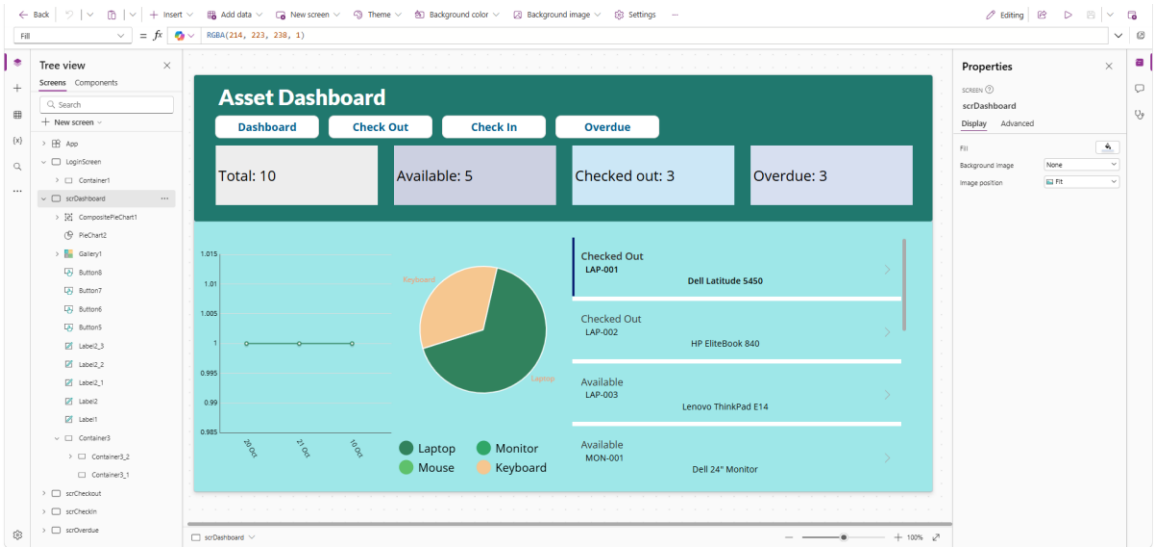
4.3 Relationship

One **Asset** has many **Checkouts** (one-to-many). A checkout is **open** while `ReturnedOn` is blank; an asset must never have more than one open checkout.

5. Canvas App Design

5.1 Screens

Screen	Purpose
Dashboard	Summary cards (Available, Checked Out, Overdue) and a list of overdue items
Check Out	Scan or select an asset → assign a user → set due date → confirm
Check In	Select a checked-out asset → confirm return
Overdue	Full list of items past their due date



5.2 Check Out screen

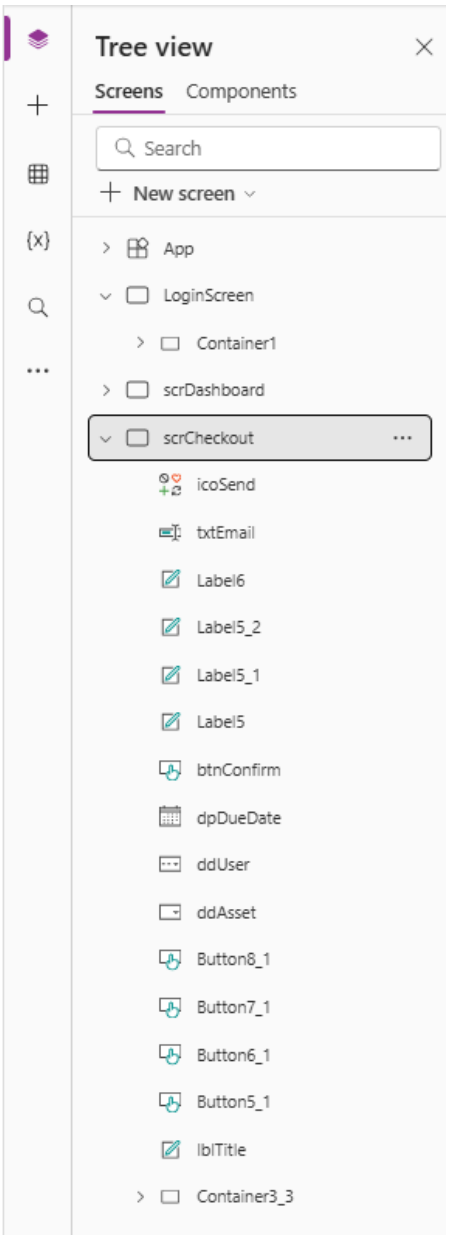
Workflow: **Scan or select asset → assign user → confirm.**

- A gallery lists assets filtered to `Status = Available`.
- A barcode scanner control can fill the Tag to find the asset.
- A combo box selects the user and a date picker selects the due date.
- The **Confirm** button runs the check-out logic.

Example logic (adjust names to match your own tables and columns):

```
If(
    // Guard: asset must still be available
    galAssets.Selected.Status <> 'Status (Assets)'.Available,
    Notify("This asset is already checked out.", NotificationType.Error),
    Patch(
        Checkouts,
        Defaults(Checkouts),
        {
            Asset: galAssets.Selected,
            User: cmbUser.Selected,
            From: Now(),
            Due: dpDue.SelectedDate
        }
    );
    Patch(
        Assets,
        galAssets.Selected,
        { Status: 'Status (Assets)'.'Checked Out' }
    );
    Notify("Check-out successful.", NotificationType.Success)
)
```

In this project the check-out is also supported by the Power Automate flow **Flow_CheckoutAsset** (see section 6.1), which performs the server-side validation.



5.3 Check In screen

The user selects an asset that is currently checked out. The app finds the open checkout record and stamps the return date, then frees the asset:

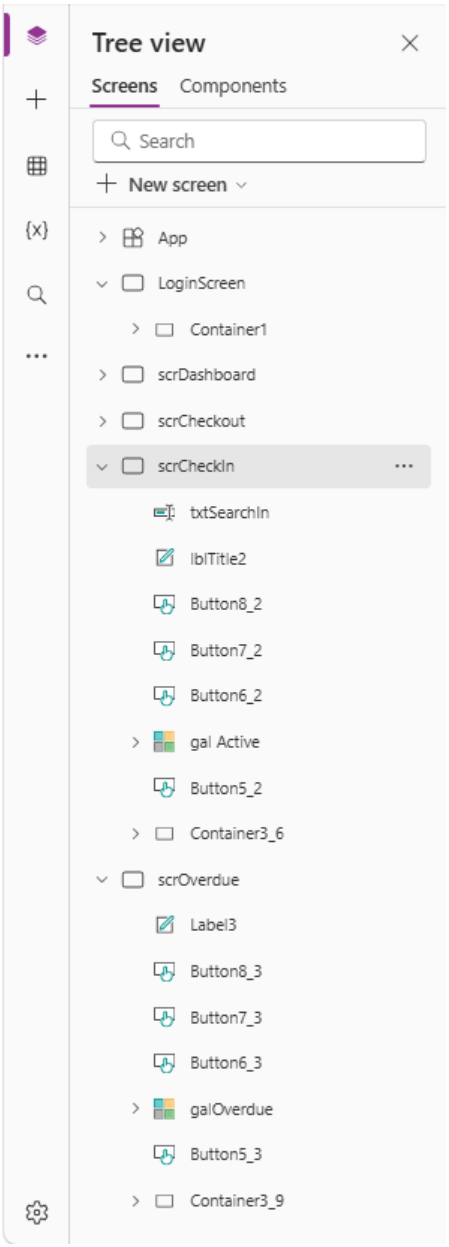
```
With(
    { openCo: LookUp(Checkouts, Asset.Tag = galOut.Selected.Tag &&
IsBlank(ReturnedOn)) },
    Patch(Checkouts, openCo, { ReturnedOn: Now() });
    Patch(Assets, galOut.Selected, { Status: 'Status (Assets)'.Available });
    Notify("Asset returned.", NotificationType.Success)
)
```

5.4 Overdue screen and dashboard

Overdue items are open checkouts whose due date has passed:

```
Filter(Checkouts, IsBlank(ReturnedOn) && Due < Now())
```

The dashboard counts use `CountRows()` over the same filter and over assets grouped by status.



6. Power Automate Flows

6.1 Flow_CheckoutAsset

- **Trigger:** Power Apps (V2) with inputs for asset, user and due date.
- **Step 1:** List rows in Checkout where the asset matches and `ReturnedOn` is blank.

- **Step 2:** Condition – if any open checkout exists, respond to the app with an error (prevents double check-out).
- **Step 3:** Otherwise add a new Checkout row and update the Asset status to *Checked Out*.
- **Step 4:** Respond to the app with success.

[INSERT SCREENSHOT: Flow_CheckoutAsset overview] [INSERT SCREENSHOT: successful test run]

6.2 Reminder flow (24 hours before due)

- **Trigger:** Recurrence – runs every hour (or once daily).
- **Step 1:** List rows from Checkout where `ReturnedOn` is blank and `Due` is between now and now + 24 hours.
- **Step 2:** Apply to each row.
- **Step 3:** Post a message in Teams (chat with the assignee) that @mentions the user, giving the asset name, tag and due date.
- **Optional:** add a `ReminderSent` yes/no column so each person is reminded only once.

[INSERT SCREENSHOT: reminder flow overview] [INSERT SCREENSHOT: Teams message received by the assignee]

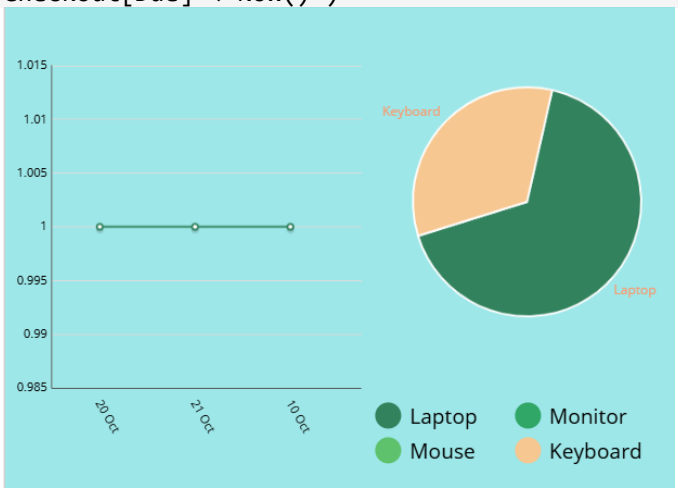
7. Analytics (Power BI)

Report	Measure	Visual
Overdue trend	Count of overdue checkouts by week or month	Line chart
Current status	Assets per Status	Donut chart

Key measures:

```
Utilisation % = DIVIDE( CALCULATE(COUNTROWS(Asset), Asset[Status] = "Checked Out"),
COUNTROWS(Asset) )

Overdue Count = CALCULATE( COUNTROWS(Checkout), ISBLANK(Checkout[ReturnedOn]),
Checkout[Due] < NOW() )
```



8. Stretch Goal – AI Builder Object Detection

An AI Builder object detection model can be trained with labelled photos of each asset type (laptop, mouse, keyboard, monitor). In the Check Out screen the user takes a photo and the model suggests the asset type, which pre-filters the asset gallery. The scanner integration is unavailable in the current setup, and the Power Apps Developer Plan does not support the required AI Builder features, prompting a request to purchase an additional plan.

9. Testing and Acceptance

#	Test case	Expected result	Actual result	Result
1	Check out an available laptop	Checkout row created; asset status = Checked Out	Checkout row created and asset marked Checked Out	Pass
2	Try to check out the same laptop again	Blocked with an error message	Blocked; error message shown	Pass
3	Check in the laptop	ReturnedOn filled; asset status = Available	ReturnedOn stamped; asset marked Available	Pass
4	Check out an item due in under 24 hours	Teams reminder with @mention arrives	Reminder received in Teams with @mention	Pass
5	Leave an item past its due date	Appears on the Overdue screen and dashboard	Item listed on Overdue screen and dashboard	Pass
6	Power chat report	Utilization and overdue trend show correct figures	Figures matched the Dataverse data	Pass

All six test cases passed, and both acceptance criteria were met: no asset could be checked out twice at the same time, and the overdue reminders fired as designed.

10. Challenges and Learnings

- Understanding one-to-many relationships and lookup columns in Dataverse.
- Using `Patch()` to write to two related tables consistently.
- Preventing double check-out with both a client-side check and a server-side flow check.
- Scheduling reminders and mentioning users in Teams.

11. Future Enhancements

- Approval step before issuing high-value assets.
- Email escalation to the manager for items overdue by more than 3 days.
- QR-code labels for every asset tag.
- Full AI Builder photo recognition.

12. Conclusion

The Asset Checkout System was completed and tested. It gives a simple, reliable way to track temporary loans of laptops and peripherals. Dataverse holds the structured data, the canvas app handles check-out and check-in, Power Automate provides validation and reminders, and Power BI gives insight. The project met both acceptance criteria: duplicate check-outs are prevented and reminders are sent before items become overdue.